

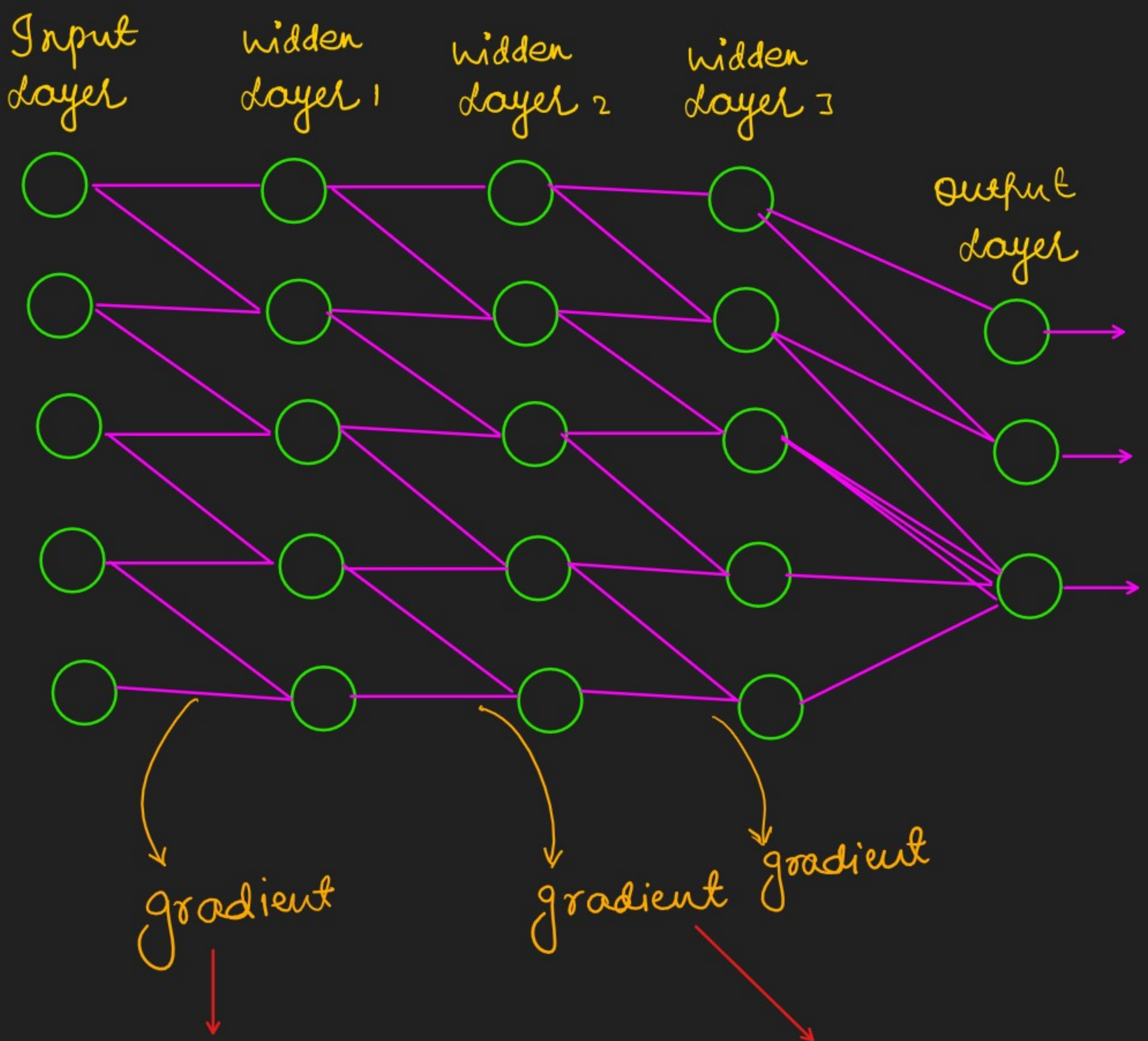
# Lecture - 20: Layer Normalization

① Training deep neural network with many layers can be challenging due to

vanishing/  
exploding  
gradient

② This leads to unstable training dynamics.

③ Layer normalization improves the stability and efficiency of neural network training.



Depends on layer output

If layer output is too large or small, gradient magnitude can become too large or small.

This affects training.

Batch normalization keep gradient stable

As the training proceeds, inputs to each layer can change (Internal covariate shift)

This delays convergence

Layer normalization prevents this



- ④ Main Idea: Adjust output of neural network to have mean zero and variance of one.

$$\begin{matrix} \xrightarrow{x_1} & \xrightarrow{x_2} & \xrightarrow{x_3} & \xrightarrow{x_4} \\ [1.1, 0.8, 2.3, 4.4] \end{matrix}$$

$$\mu = \frac{1.1 + 0.8 + 2.3 + 4.4}{4} = 2.15$$

$$\begin{aligned} \text{var} &= \frac{1}{4} \left( (1.1 - 2.15)^2 + (0.8 - 2.15)^2 \right. \\ &\quad \left. + (2.3 - 2.15)^2 + (4.4 - 2.15)^2 \right) \\ &= 2.0025 \end{aligned}$$

$$\begin{aligned} \text{Normalized: } & \left[ \frac{x_1 - \mu}{\sqrt{\text{var}}} \quad \frac{x_2 - \mu}{\sqrt{\text{var}}} \quad \frac{x_3 - \mu}{\sqrt{\text{var}}} \quad \frac{x_4 - \mu}{\sqrt{\text{var}}} \right] \\ & : [0.74, -0.95, 0.1, 1.59] \end{aligned}$$

- ⑤ In GPT-2 and modern transfer architecture, layer normalization is typically applied before and after the multi-head attention module and before the final output layer.

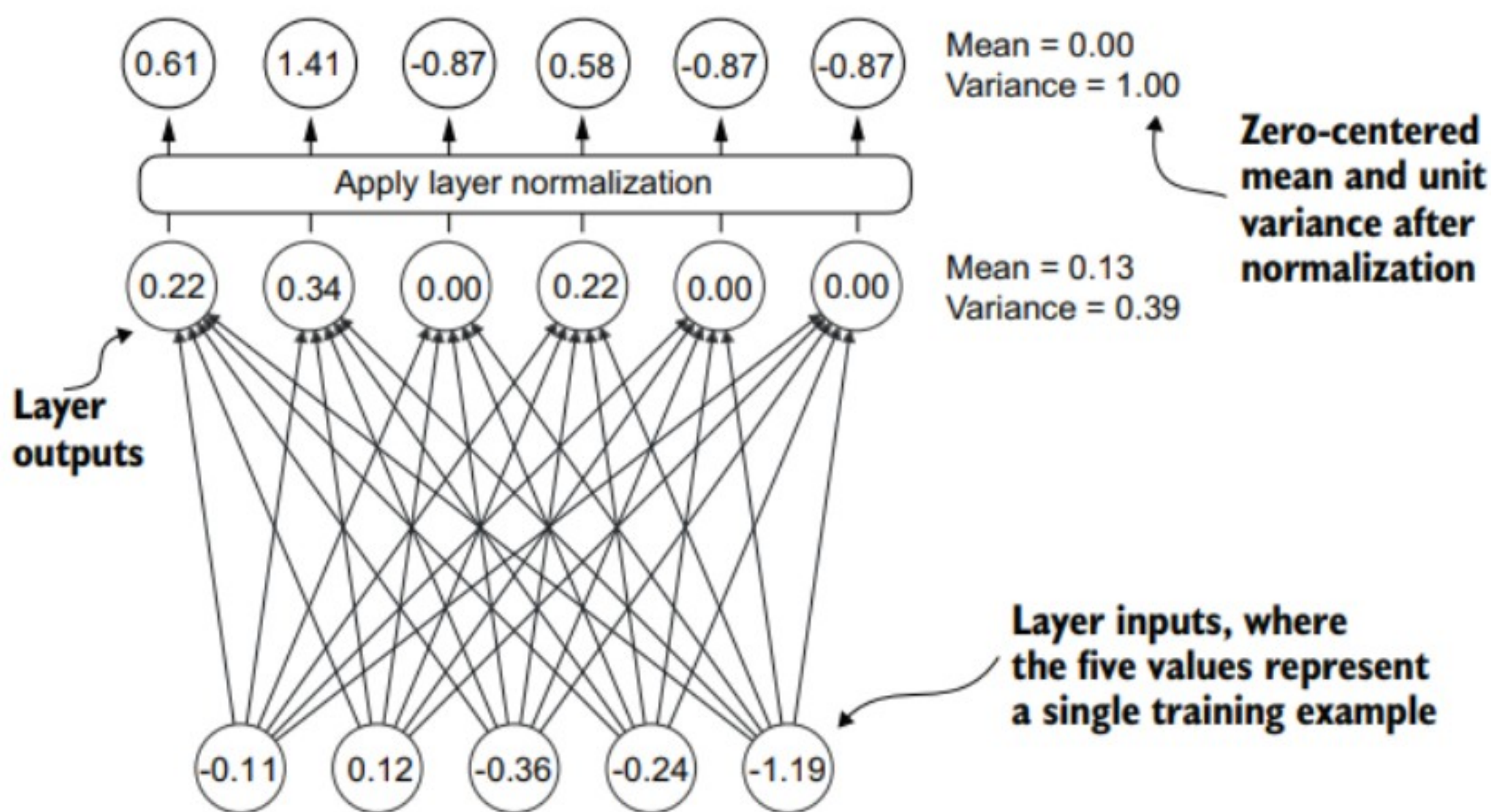
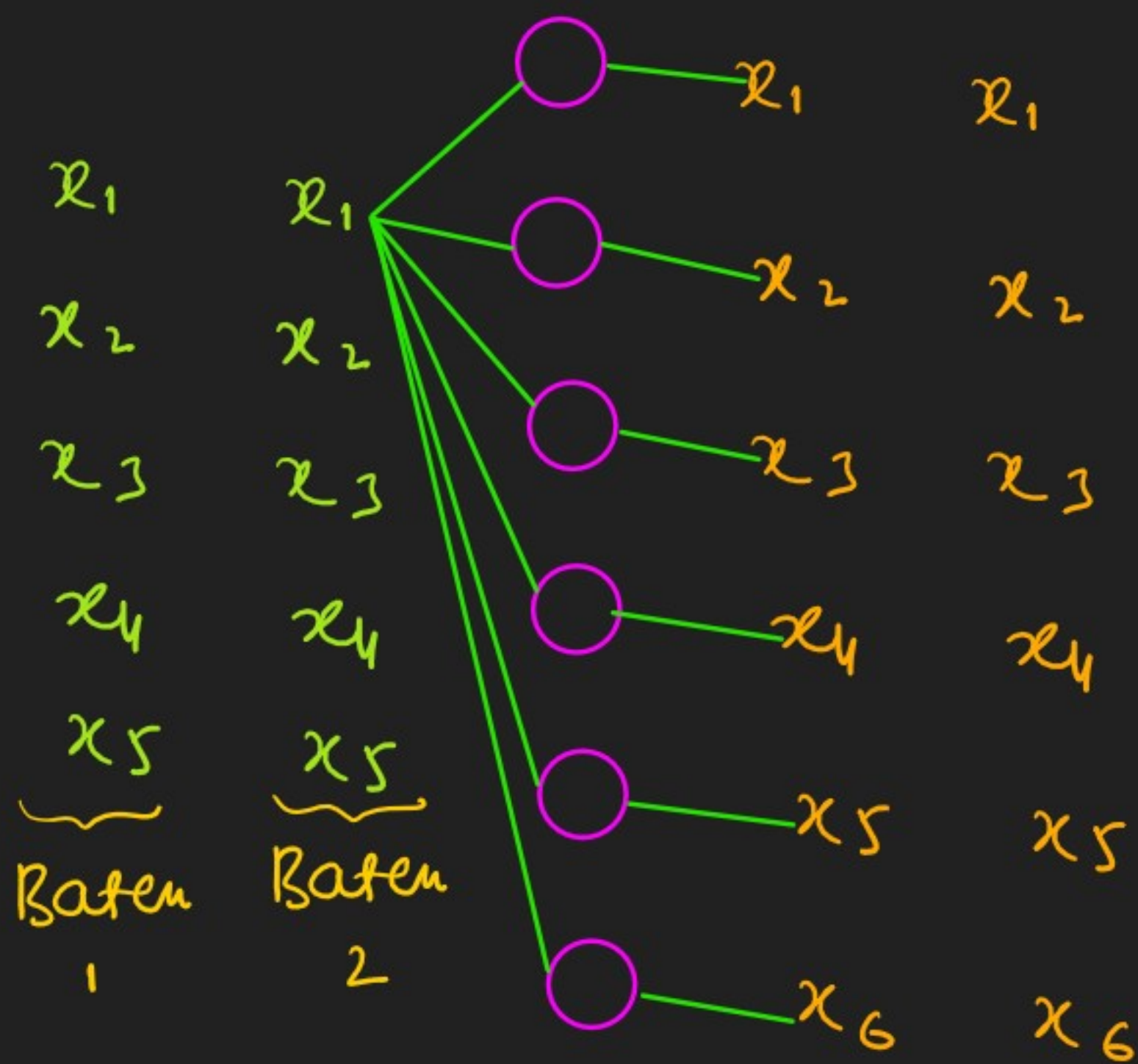


Figure 4.5 An illustration of layer normalization where the six outputs of the layer, also called activations, are normalized such that they have a 0 mean and a variance of 1.



⑥ let's move to code



⑦ Layer vs normalization

Normalize  
along feature dimension  
independently of  
batch size

Available  
hardware  
dictates  
batch size

leads to more  
flexibility and  
stability for  
distributed  
training/environ-  
ment which  
lack resources.